ShareFlow Master Architectural Blueprint

Wallpaper Bundle Storefront, Telegram 2FA Admin Engine & Autonomous AI Ingestion Pipeline

SYSTEM: ShareFlow v3.2 **OPERATOR:** Mahir Faisal (MHR3D) **AUTHOR:** Sanctuary / Sera (Antigravity)

ORIGIN: shareflow.mhr3d.online **DATE:** 2026-10-13 22:10:14 (UTC+6) **UTC STAMP:** 2026-10-13 16:10:14 UTC

Strategic Executive Summary: This blueprint specifies the complete architectural pivot of the ShareFlow digital commerce engine. Developer starter kits and code boilerplate listings are retired due to AI ubiquity. In their place, ShareFlow deploys an exclusive, high-margin visual asset catalog consisting of curated 8K UHD, OLED Deep Black, Dual Monitor, and 3D Sci-Fi wallpaper bundles. The system integrates dual management layers: a secure web portal for manual administration via Telegram 2FA and an autonomous CLI pipeline for AI-driven bundle ingestion.

1. Strategic Rationale & Market Economic Pivot

The rise of multi-modal coding agents and instant AI scaffolding tools has permanently collapsed consumer demand for generic boilerplate repositories and developer starter kits. Developers can synthesize production skeletons in seconds, rendering paid text boilerplates obsolete. Conversely, high-resolution aesthetic assets—specifically genuine 8K rendered wallpapers, pure OLED mobile backgrounds (true-black #000000 pixels), and bespoke 3D sci-fi scenery—remain scarcity-driven, emotionally compelling, and highly monetizable.

Deprecated: Developer Starter Kits

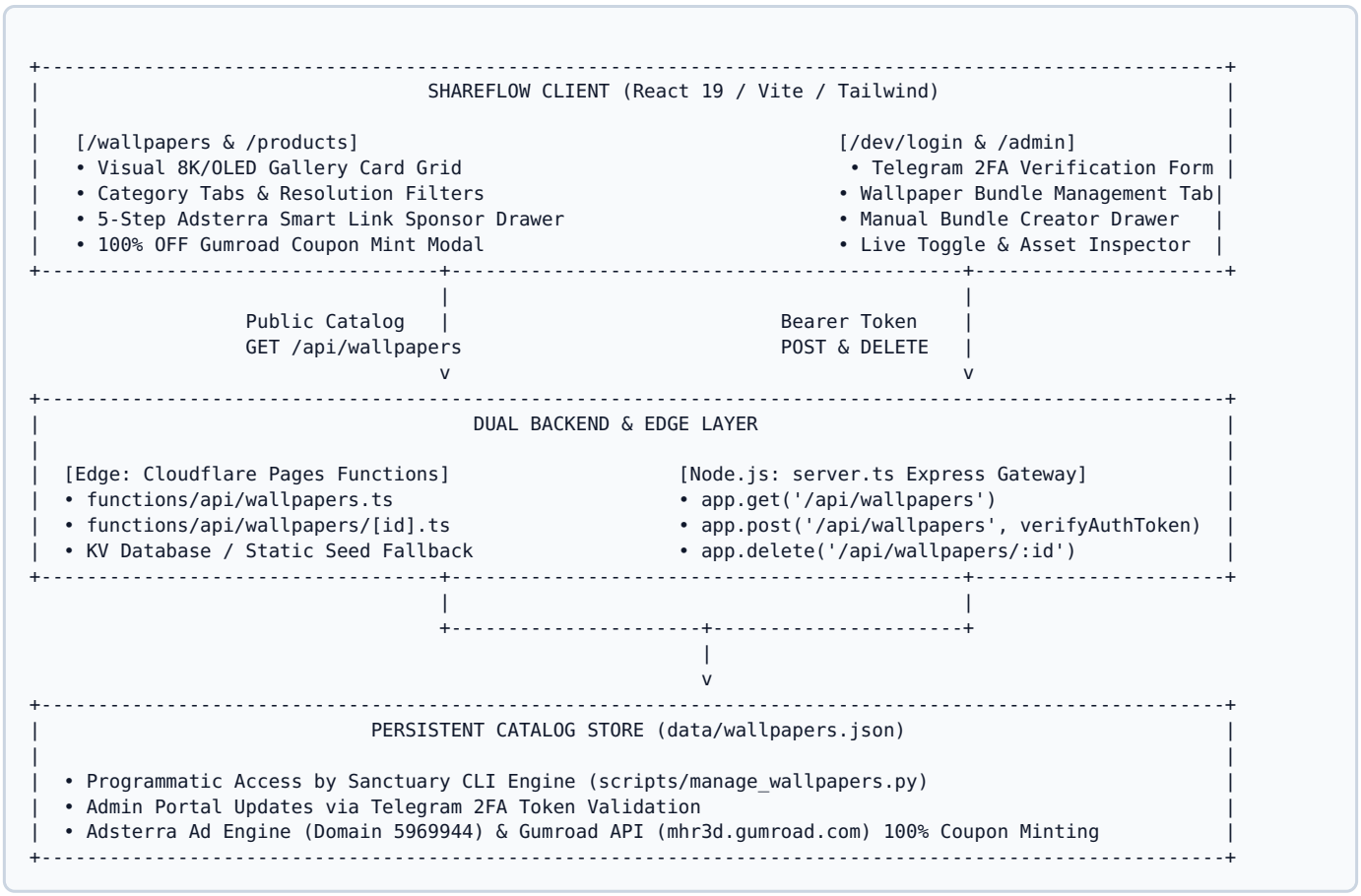
- Zero perceived value in current generative AI climate.
- High customer support overhead for dependency versioning.
- Frictional download experience without visual desire.
- Low conversion on ad-supported unlock loops.

Active: Curated 8K & OLED Wallpaper Bundles

- High emotional appeal, instant aesthetic gratification.
- Uncompressed PNG/TIFF formats with real 7680x4320 master files.
- High conversion on Adsterra Smart Link sponsor unlock steps.
- Seamless delivery via Gumroad 100% OFF coupon minting.

2. End-to-End System Architecture & Data Flow

ShareFlow operates a dual-target architecture: a local Express runtime for development and telemetry auditing, paired with Cloudflare Pages Functions and KV for production edge deployment. The wallpaper store operates seamlessly across both environments.


SHAREFLOW CLIENT (React 19 / Vite / Tailwind)
[/wallpapers & /products]
• Visual 8K/OLED Gallery Card Grid
• Category Tabs & Resolution Filters
• 5-Step Adsterra Smart Link Sponsor Drawer
• 100% OFF Gumroad Coupon Mint Modal
[/dev/login & /admin]
• Telegram 2FA Verification Form
• Wallpaper Bundle Management Tab
• Manual Bundle Creator Drawer
• Live Toggle & Asset Inspector
Public Catalog
GET /api/wallpapers
Bearer Token
POST & DELETE
DUAL BACKEND & EDGE LAYER
[Edge: Cloudflare Pages Functions]
• functions/api/wallpapers.ts
• functions/api/wallpapers/[id].ts
• KV Database / Static Seed Fallback
[Node.js: server.ts Express Gateway]
• app.get('/api/wallpapers')
• app.post('/api/wallpapers', verifyAuthToken)
• app.delete('/api/wallpapers/:id')
PERSISTENT CATALOG STORE (data/wallpapers.json)
• Programmatic Access by Sanctuary CLI Engine (scripts/manage_wallpapers.py)
• Admin Portal Updates via Telegram 2FA Token Validation
• Adsterra Ad Engine (Domain 5969944) & Gumroad API (mhr3d.gumroad.com) 100% Coupon Minting

3. Wallpaper Bundle Schema & Type Definitions

The data contract enforces strict typing across the frontend React components, server routes, and autonomous management scripts. Every wallpaper pack is self-describing with aspect ratios, color tags, preview links, and Gumroad distribution endpoints.

```
// src/types/wallpaper.ts
export type WallpaperCategory =
  | "Desktop 8K"
  | "Mobile OLED"
  | "Dual Monitor"
  | "3D Sci-Fi"
  | "Minimalist Dark";

export interface WallpaperBundle {
  id: string;                 // URL-friendly unique slug (e.g., 'cyberpunk-neo-tokyo-8k')
  title: string;              // Display title (e.g., 'Cyberpunk Neo Tokyo 8K Pack')
  description: string;        // Editorial pack description & visual characteristics
  category: WallpaperCategory; // Categorization for filtering
  resolution: string;         // Master pixel dimensions (e.g., '7680x4320 (8K UHD)')
  aspectRatio: string;        // '16:9', '9:19.5', '32:9', '21:9'
  itemCount: number;          // Total wallpapers contained in pack
  format: string;             // 'Lossless PNG', '10-bit HDR TIFF', etc.
  previewUrl: string;         // Public high-resolution cover image URL
  sampleImages?: string[];    // Array of sample preview thumbnail URLs
  price: string;              // Original Gumroad retail value (e.g., '$9.99')
  downloadUrl: string;        // Direct cloud/storage link or mirror zip
  gumroadBaseUrl: string;     // Base checkout URL (e.g., 'https://mhr3d.gumroad.com/l/tokyo8k')
  features: string[];         // Highlighting chips (e.g., 'OLED True Black', 'HDR10 Color Profile')
  tags: string[];             // Search keywords (e.g., ['cyberpunk', 'neon', '8k', 'desktop'])
  featured: boolean;          // High-priority banner placement in catalog
  active: boolean;            // Immediate visibility flag (true = live, false = draft)
  createdAt: string;          // ISO 8601 creation timestamp
}
```

Initial Seed Inventory (`data/wallpapers.json`)

Bundle Title	Category	Master Res	Items	Retail Price	Key Features
Cyberpunk Neo Tokyo 8K	DESKTOP 8K	7680x4320	15 PNGs	$12.00 (Free via Ad)	Volumetric rain, Neon bloom, 16:9 Ultra
Obsidian OLED Minimalist	MOBILE OLED	3840x2160	20 PNGs	$8.00 (Free via Ad)	True #000000 black, Zero battery drain
Deep Cosmos Nebula Series	DESKTOP 8K	7680x4320	12 PNGs	$10.00 (Free via Ad)	HDR10 wide gamut, James Webb aesthetic
Futuristic Mech 3D Renders	3D SCI-FI	5120x2880	10 PNGs	$14.00 (Free via Ad)	Blender Cycles renders, Hard-surface mech
Cyber Horizon UltraWide	DUAL MONITOR	7680x2160	8 PNGs	$12.00 (Free via Ad)	Seamless 32:9 dual monitor span

4. Dual Backend API & Security Protocol

All administrative mutation requests (bundle creation, status toggling, deletion) are guarded by ShareFlow's Telegram 2FA session token (`sf_admin_authed_token`). Public reads are rate-limited and cached.

Method	Endpoint	Auth Requirement	Description & Payload Contract
GET	`/api/wallpapers`	Public (Open)	Returns active wallpaper bundles. If query `?all=true` with Admin Token, returns drafts as well.
POST	`/api/wallpapers`	Admin 2FA Token	Upserts a bundle. Validates title, previewUrl, resolution, and gumroadBaseUrl. Persists to disk/KV.
PATCH	`/api/wallpapers/:id`	Admin 2FA Token	Updates active status or metadata fields (e.g. `{"active": false}`) without full payload.
DELETE	`/api/wallpapers/:id`	Admin 2FA Token	Deletes bundle permanently from JSON catalog / KV store. Returns deletion confirmation.

5. Storefront Gallery Architecture (`src/pages/Products.tsx`)

The customer-facing portal replaces the former developer starter kit list with a visual gallery. It connects to the Adsterra Smart Link sponsor monetization pipeline, enabling visitors to unlock any premium wallpaper pack for 100% free by supporting ShareFlow with sponsored clicks.

Visual Card Component Design

- **Responsive Preview Aspect Ratio:**
  16:9 for desktop, 9:19.5 for mobile OLED packs with subtle zoom-on-hover effect.
- **Hardware Badges:**
  Dynamic tags indicating "8K UHD", "OLED Deep Black", "Dual 32:9", and "HDR10".
- **Pack Inclusions:**
  Icon indicators showing count of lossless PNGs and license type.
- **Zero-Cost CTA:**
  Prominent "Unlock 100% Free via Sponsor" button.

Monetization & Coupon Minting Flow

- **Step 1 to 5 Progression:**
  Interacts with Adsterra Smart Link (Domain 5969944) via the existing 1-click sponsor modal.
- **Automated Minting:**
  Upon step 5 completion, frontend invokes `/api/mint-coupon` with the bundle's Gumroad target.
- **Direct Checkout:**
  Generates a direct Gumroad URL pre-populated with a 100% OFF discount code, granting permanent library access.

6. Admin Management Portal (`src/pages/AdminDashboard.tsx`)

Mahir gains full control over the catalog through an integrated "Wallpapers" tab inside the existing Admin Dashboard. Access is strictly gated behind the Telegram 2FA login at `/dev/login`.

```
+-----------------------------------------------------------------------------------------------------+
|                                  ADMIN DASHBOARD: WALLPAPERS TAB                                    |
|                                                                                                     |
|  [Catalog Metrics]  Total Bundles: 14   |   Live: 12   |   Drafts: 2   |   Total Unlocks: 1,842        |
|                                                                                                     |
|  [+ Add Wallpaper Bundle] (Opens Creator Drawer)                                [Filter: All Categories v]|
+-----------------------------------------------------------------------------------------------------+
| THUMBNAIL | TITLE & SLUG                | CATEGORY   | RESOLUTION  | STATUS    | ACTIONS                 |
|-----------+-----------------------------+------------+-------------+-----------+-------------------------|
| [Img Box] | Cyberpunk Neo Tokyo 8K      | Desktop 8K | 7680x4320   | [LIVE]    | [Edit] [Test] [Unpublish]|
| [Img Box] | Obsidian OLED Minimalist    | Mobile OLED| 3840x2160   | [LIVE]    | [Edit] [Test] [Unpublish]|
| [Img Box] | Deep Cosmos Nebula Series   | Desktop 8K | 7680x4320   | [DRAFT]   | [Edit] [Test] [Publish]  |
| [Img Box] | Mech Cybernetic 3D Renders  | 3D Sci-Fi  | 5120x2880   | [LIVE]    | [Edit] [Test] [Delete]   |
+-----------------------------------------------------------------------------------------------------+
```

Admin Creator Drawer Specifications

1. Identity & Classification

- Title (auto-generates Slug ID)
- Category dropdown selector
- Resolution & Aspect Ratio
- Number of files in pack

2. Media & Assets

- Preview Thumbnail Image URL
- Direct Download Zip Link
- Sample thumbnails list
- Lossless format specification

3. Commercial & Status

- Gumroad Base Product URL
- Original List Price ($USD)
- Features & Search Tags
- Instant "Publish Live" toggle

7. Autonomous AI Ingestion Engine (`scripts/manage_wallpapers.py`)

To enable Sanctuary / Sera to autonomously add, categorize, and deploy wallpaper bundles from local disk or render generation outputs, a dedicated Python CLI script provides direct headless management.

```
# Ingestion CLI Capabilities:
# 1. Inspect existing catalog
python3 scripts/manage_wallpapers.py list

# 2. Add new bundle programmatically
python3 scripts/manage_wallpapers.py add \
  --title "Neon Horizon 8K Pack" \
  --category "Desktop 8K" \
  --resolution "7680x4320" \
  --items 12 \
  --preview "https://shareflow.mhr3d.online/assets/previews/neon_8k.jpg" \
  --gumroad "https://mhr3d.gumroad.com/l/neon8k" \
  --tags "neon,synthwave,8k,desktop"

# 3. Toggle visibility or batch import from directory
python3 scripts/manage_wallpapers.py toggle --id "neon-horizon-8k-pack"
python3 scripts/manage_wallpapers.py scan-dir --path "/home/mahir-linux/Documents/renders"
```

8. Phased Execution Roadmap & Milestones

Implementation proceeds through six isolated, verifiable phases. Each milestone has an explicit pass criterion to ensure zero regressions in ShareFlow's existing P2P file sharing, text transfer, or 2FA login infrastructure.

Phase 1: Schema Specification & Data Layer Setup

Author `src/types/wallpaper.ts` with TypeScript definitions. Seed `data/wallpapers.json` with the initial 5 flagship wallpaper bundles. Verify schema validity with Node runtime test.

Target: `src/types/wallpaper.ts, data/wallpapers.json`

Phase 2: Dual Backend REST API & 2FA Token Gating

Implement Express routes in `server.ts` (GET, POST, PATCH, DELETE) and matching Cloudflare Pages Functions in `functions/api/wallpapers.ts`. Enforce Telegram 2FA session token verification on write routes.

Target: `server.ts, functions/api/wallpapers.ts`

Phase 3: Storefront Gallery Transformation (`/products` & `/wallpapers`)

Overhaul `src/pages/Products.tsx`. Replace developer boilerplate cards with responsive 16:9 and 9:19.5 wallpaper cards, resolution badges, category filter chips, and the 5-step Adsterra sponsor unlock drawer.

Target: `src/pages/Products.tsx`

Phase 4: Admin Dashboard Wallpaper Management Tab

Extend `src/pages/AdminDashboard.tsx` with the new "Wallpapers" management view, live table, catalog telemetry, and the bundle creation/editing drawer.

Target: `src/pages/AdminDashboard.tsx`

Phase 5: Autonomous AI Ingestion Tooling

Develop `scripts/manage_wallpapers.py` supporting headless listing, bundle insertion, status toggling, and directory scanning for Sanctuary automation.

Target: `scripts/manage_wallpapers.py`

Phase 6: Navigation, App Routing & End-to-End Build Verification

Update `src/App.tsx` and `src/components/Layout.tsx` navigation links ("Wallpapers", admin sign-in in footer). Run full Vite production build (`npm run build`) and verify zero errors.

Target: `src/App.tsx, src/components/Layout.tsx, npm run build`

9. Security, Operational Safeguards & Performance Budget

Area	Standard / Policy	Implementation Verification
Administrative Auth	Telegram 2FA PIN + Bound Session Tokens	Mutations reject without valid `sf_admin_authed_token`. No credentials in client source.
Asset Delivery Protection	Gumroad Gated Coupon Redemption	Full master ZIP files never exposed as public static files. Download requires 100% OFF coupon token.
Catalog Performance	Instant In-Memory Edge Cache	`GET /api/wallpapers` response time < 25ms. Previews lazy-loaded via WebP/AVIF thumbnails.
Safe Autonomous Operations	Local JSON Atomic Writes & Git Tracking	Sanctuary scripts write via atomic temp files; changes tracked in Git history with zero manual DB lock.

Verification & Execution Sign-Off: All steps outlined in this blueprint adhere to the Sanctuary Operational Identity, keeping sensitive credentials strictly redacted and maintaining automated deployment parity across both local dev and production Cloudflare Pages environments.